

UNIVERSITY OF EDINBURGH
SCHOOL OF INFORMATICS

INFR11287 ADVANCED TOPICS IN NATURAL LANGUAGE
PROCESSING

Mock Examination Paper 2

Adaptation, Reasoning, Long Context, Summarisation, and Safety

Time allowed: 2 hours

INSTRUCTIONS TO CANDIDATES

1. This is a practice paper, not an official examination paper.
2. Answer all questions. Different questions may have different numbers of total marks.
3. The paper is worth 50 marks in total.
4. Notes, calculators, and dictionaries are not permitted in the real examination.
5. Please answer concisely. Unless otherwise stated, do not write more than one or two sentences per mark assigned.

1. A legal technology company wants to build an LLM system that summarises long contracts and answers follow-up questions about obligations, deadlines, and penalties. It has a small set of expert-written summaries, a larger set of unlabelled contracts, and a budget that is not large enough for repeated full fine-tuning of a 70B model.
 - (a) Define the summarisation task in this setting. In your answer, distinguish extractive and abstractive summarisation, and explain one risk of using a purely abstractive model for legal text. [3 marks]
 - (b) The company considers instruction fine-tuning, RLHF-style preference optimisation, and the coursework-style use of verifiable rewards for reasoning tasks. Compare these approaches for this application, mentioning one benefit and one risk or limitation. [4 marks]
 - (c) The company decides to use LoRA. Write the basic low-rank update equation for a weight matrix W , and explain how this can be applied to the query, key, and value matrices of a Transformer. State one training-cost advantage over full fine-tuning. [4 marks]
 - (d) A stronger proprietary model is available through an API. Explain how knowledge distillation could be used to improve the smaller in-house model, and note one risk of this distillation process. [2 marks]
 - (e) Describe an evaluation plan for the legal summarisation system. Your answer should include one automatic metric, one human evaluation component, and one failure mode that the evaluation must explicitly check. [2 marks]

2. The same company later wants the model to read an entire 80-page contract in one prompt. Some clauses relevant to a question may appear near the beginning, some in the middle, and some near the end. The engineering team is considering long-context models, retrieval, and decoding-time reasoning methods.
- (a) Explain why Transformer models need positional encodings. Then compare sinusoidal positional encodings and RoPE in terms of how they represent position. [4 marks]
 - (b) RoPE-based models often require context-extension methods. Explain the intuition behind positional interpolation or YaRN, and state one tradeoff involved in extending the context window. [3 marks]
 - (c) Full attention becomes expensive for very long inputs. Explain why its cost grows quickly with sequence length, and describe one efficient-attention strategy that could reduce cost. [2 marks]
 - (d) Long context does not guarantee that the model uses all evidence equally well. Describe two ways you would evaluate whether the model can actually use information in the middle of a long contract. [3 marks]
 - (e) The team considers chain-of-thought prompting and self-consistency for clause-based reasoning. Explain how self-consistency works, and give one reason it might improve answers and one reason it might be unsuitable for deployment. [3 marks]

3. The following are multiple choice questions. For each question, choose the answer that is the most correct. Each question is worth 2 marks. [20 marks]

- (a) A core weakness of ROUGE for summarisation evaluation is that:
 - A. It cannot be calculated on short summaries.
 - B. It mainly measures lexical overlap and can miss semantic equivalence.
 - C. It always rewards hallucinated content.
 - D. It requires access to model parameters.
- (b) A pointer-generator network in abstractive summarisation is useful because it:
 - A. Prevents the model from copying any source token.
 - B. Can mix generating from a vocabulary with copying from the source.
 - C. Replaces attention with a bag-of-words model.
 - D. Is only used for extractive summarisation.
- (c) RoPE is usually applied to:
 - A. The value vectors only.
 - B. The query and key vectors used to compute attention scores.
 - C. The tokeniser vocabulary.
 - D. The final softmax layer only.
- (d) Position interpolation for long context mainly tries to:
 - A. Remove all position information from the model.
 - B. Map longer position indices into a range more familiar from pre-training.
 - C. Replace all attention with convolution.
 - D. Increase the number of parameters in every layer.
- (e) In self-consistency prompting, the model usually:
 - A. Samples multiple reasoning paths and aggregates the final answers.
 - B. Uses exactly one greedy chain-of-thought and never samples.
 - C. Fine-tunes all model weights using human preferences.
 - D. Removes all intermediate reasoning steps from the prompt.
- (f) A Shapley value analysis of reasoning steps estimates:
 - A. The average marginal contribution of a step across possible subsets/orderings.
 - B. The number of tokens in a prompt.
 - C. The BLEU score of a translation system.
 - D. The perplexity of a language model on Wikipedia.
- (g) In knowledge distillation, the student model is trained to:
 - A. Always be larger than the teacher model.
 - B. Imitate useful behaviour or outputs of a teacher model.
 - C. Delete all teacher-generated examples after training.
 - D. Use only hand-written grammar rules.
- (h) The main training-time advantage of PEFT over full fine-tuning is that:
 - A. It removes the need for data.
 - B. It trains fewer task-specific parameters and usually requires less memory.
 - C. It prevents overfitting in all settings.
 - D. It cannot be combined with instruction tuning.
- (i) A core risk of RLHF or reward-based optimisation is:

- A. The model may optimise the reward signal without improving the true target behaviour.
 - B. It can only be used for machine translation.
 - C. It prevents human preference data from being used.
 - D. It removes the need to evaluate the resulting model.
- (j) BPE-dropout is best understood as:
- A. Randomly dropping possible merge operations to expose the model to varied segmentations.
 - B. Removing all rare words from the vocabulary.
 - C. A method for evaluating translation adequacy.
 - D. A watermarking detector for generated text.